

Retail Inventory and Stock Management System - Test Plan | Team E10

SOFTWARE ENGINEERING MINI PROJECT

RETAIL INVENTORY AND STOCK MANAGEMENT SYSTEM

TEST PLAN

Prepared in IEEE 829-1998 style

Document ID: RIMS-TP-E10-v1.0

Version: 1.0

Date: 29 September 2026

Course: Software Engineering | Academic Year: 2026–27

Implementation: C / C++ console application with file-based storage

Name	SRN	Role
Malavika A	PES1UG24CS257	Team member
Manoj Kumar	PES1UG24CS264	Team member
Mayank M	PES1UG24CS266	Team member
LG Vivyn Krishnaa	PES1UG24CS248	Team member

Revision History

Version	Date	Description	Author
1.0	29 Sep 2026	Initial test plan aligned with Phase-1 SRS, IEEE-style sections, security validation and SRS traceability	Team E10

1. Test Plan Identifier

RIMS-TP-E10-v1.0. This is the master test plan for the Retail Inventory and Stock Management System. It is derived from the Phase-1 Requirements Analysis and Specification supplied for Team E10 and covers functional, validation, persistence, integration, system and security-validation testing of the C/C++ file-based console application.

2. Introduction

2.1 Purpose

The purpose of this Test Plan is to define the scope, approach, resources, test conditions, pass/fail criteria and traceability for testing the Retail Inventory and Stock Management System. Testing shall verify that every functional requirement and verifiable functional requirement in the Phase-1 SRS is covered by at least one planned test case.

2.2 Scope

Testing covers FR01–FR14 and VFR01–VFR09 from the supplied SRS. The scope includes user access, product creation/update/removal, product search and display, stock adjustment, stock availability, stock movement history, inventory listing, inventory summary and low-stock identification, file storage, and loading of saved data at application startup.

2.3 References

- SRS: Retail Inventory and Stock Management System – Phase 1, Team E10
- Requirements Traceability Matrix (RTM) and Use Cases UC01–UC11 from the Phase-1 SRS
- IEEE Std 829-1998 Standard for Software Test Documentation

2.4 Definitions

TC = Test Case; UT = Unit Test; IT = Integration Test; ST = System Test; SEC = Security Validation Test; FR = Functional Requirement; VFR = Verifiable Functional Requirement; RTM = Requirements Traceability Matrix. A Critical defect prevents execution or causes data loss/corruption; a Major defect prevents a required feature from working correctly; a Minor defect has limited impact and a practical workaround.

3. Test Items

Item	Description	SRS / Component	Test Level
3.1 Application Access / Startup	Entry into the application and loading of saved product and stock-history data	Application / File Manager	IT, ST, SEC
3.2 Product Record Management	Add, update, remove and display product records	Product Record Module	UT, IT, ST
3.3 Search Module	Search by product ID or name and handle unknown search keys	Search / Product Module	UT, IT, ST
3.4 Stock Adjustment Module	Increase/decrease stock and enforce available-quantity rules	Stock Module	UT, IT, ST

3.5 Stock Availability	Display current available quantity for a selected product	Inventory Module	UT, ST
3.6 Stock Movement History	Record product ID, quantity change, date/time and adjustment type; retrieve history	History Module / File Manager	UT, IT, ST
3.7 Inventory View and Summary	List products, quantities and identify low-stock products	Inventory / Summary Module	UT, ST
3.8 Input Validation	Required fields, duplicate IDs, negative price/quantity, invalid search and invalid stock operations	Validator / Input Handling	UT, ST, SEC
3.9 Data Files	Product records and stock movement history stored and retained across restart	File Storage	IT, ST, SEC

4. Software Risk Issues

ID	Risk	Impact	Likelihood	Mitigation in Testing
R-1	Incorrect product quantity or stock calculation	High	Medium	TC-09 to TC-12; verify before/after quantities and boundary values
R-2	Invalid input creates an invalid product or stock state	High	Medium	TC-02, TC-03, TC-10, TC-11, TC-21, TC-22
R-3	Product records or stock history are lost or incorrectly rewritten	High	Medium	TC-17, TC-18, TC-24
R-4	Duplicate product ID creates conflicting records	High	Medium	TC-02, TC-21
R-5	Stock-out allows quantity greater than available stock	High	Medium	TC-11, TC-22
R-6	Stock movement history is incomplete or inaccurate	Medium	Medium	TC-13, TC-14, TC-24
R-7	Low-stock threshold is evaluated incorrectly	Medium	Medium	TC-16, TC-25
R-8	Security/access validation is insufficient for the stated authorized-user requirement	High	Low	TC-19 to TC-20; only SRS-supported access and input behavior is validated

5. Features to be Tested

Features are listed directly against the requirements in the Phase-1 SRS. No additional business features are introduced.

5.1 Security Validation

Security validation is limited to security-relevant behavior that is actually supported by the supplied SRS. The SRS specifies an authorized user entry requirement (FR01), input validation requirements (VFR02 and VFR06), and file-based persistence (FR13/VFR08). Therefore, testing verifies access behavior, rejection of malformed/invalid input, prevention of invalid stock states, and integrity/retention of stored records. The SRS does not specify password hashing, encryption, role-based access, session timeout, network security, or audit logging; these are not claimed or tested as requirements.

Security validation ID	What is validated	Method	Test Case
SEC-1 (FR01)	Authorized user can enter the inventory application; invalid/unauthorized access is rejected if an access mechanism is implemented	Manual / negative access test	TC-19
SEC-2 (VFR02)	Missing required fields, negative unit price and negative quantity are rejected without creating an invalid record	Boundary / negative input	TC-03, TC-21
SEC-3 (VFR06)	Stock-out greater than available quantity is rejected and stored quantity remains valid	Negative stock test	TC-11, TC-22
SEC-4 (FR13, VFR08)	Stored product and stock-history files retain valid records across restart and are not changed by rejected operations	File inspection / persistence test	TC-17, TC-18, TC-24
SEC-5 (VFR01)	Duplicate product IDs cannot be used to create conflicting product records	Negative input / file inspection	TC-02, TC-21

Security pass criteria: no unauthorized or invalid operation may create an invalid inventory state; malformed input shall not crash the program; rejected operations shall not incorrectly alter stored product or stock-history data; and required stored records shall remain available after restart.

5.2 Functional Features

Feature	SRS Requirements	Test Cases
User access / startup	FR01, FR14	TC-18, TC-19
Add product	FR02, VFR01, VFR02	TC-01, TC-02, TC-03, TC-21
Update product	FR03	TC-04
Remove product	FR04	TC-05
Search product	FR05, VFR03, VFR04	TC-06, TC-07, TC-22
View product details	FR06	TC-08
Adjust stock	FR07, VFR05, VFR06	TC-09, TC-10, TC-11, TC-23
Check stock availability	FR08	TC-12
Record stock movements	FR09, VFR07	TC-13, TC-24

View stock movement history	FR10	TC-14
View inventory	FR11	TC-15
Inventory summary / low-stock	FR12, VFR09	TC-16, TC-25
File persistence	FR13, VFR08	TC-17, TC-24
Load saved data	FR14, VFR08	TC-18

5.3 Non-Functional Test Considerations

The supplied Phase-1 SRS does not define separate NFR identifiers. Accordingly, no independent NFRs are introduced into the traceability matrix. Basic reliability, input robustness and usability observations may be recorded during execution when they directly affect the stated FR/VFR behavior, but they are not treated as additional requirements.

6. Features Not to be Tested

- External database integration; the SRS specifies file handling without an external database.
- Cloud deployment, networking, remote synchronization and hosted services.
- Payment processing, billing, supplier management, sales processing and customer management.
- Mobile or web interfaces; the specified system is a menu-driven C/C++ application.
- Advanced forecasting, analytics, automated procurement and other functionality not specified in the SRS.
- Password hashing, encryption, role-based access control, session timeout and audit logging because these are not specified in the supplied SRS.
- Internal compiler, operating-system and standard-library implementation details.

7. Approach

Level	Approach	Tools / Technique
Unit testing	Test validation, product operations, stock calculations and file-handling functions in isolation where modular functions are available.	C/C++ assert-based harness or manual function tests
Integration testing	Verify interaction between product records, stock adjustment, history storage and file persistence.	Test driver programs, seeded data files
System testing	Execute TC-01 to TC-25 on the complete console application using the normal menu flow.	Manual execution and recorded test data
Security validation	Perform access, malformed-input, duplicate-ID, insufficient-stock and file-integrity checks described in Section 5.1.	Manual negative tests and file inspection
Regression testing	Re-run affected and previously passed test cases after each correction or change.	Test-case checklist / script where available
Boundary testing	Test zero, negative, duplicate, missing, excessive and threshold-related values specified by the SRS.	Boundary-value and equivalence-partition testing

Traceability aspect: every planned test case identifies the SRS requirement(s) it verifies. Every FR01–FR14 and VFR01–VFR09 appears in Section 18, and Section 19 maps each requirement to its test cases.

8. Item Pass / Fail Criteria

- A test case passes when the actual result matches the expected result and the associated SRS requirement is satisfied.
- A test case fails when the actual result differs from the expected result, an invalid operation is accepted, required information is missing, or stored data is incorrectly changed/lost.
- A requirement is considered verified when all linked test cases pass.
- The build is accepted for the planned test cycle when all mandatory test cases pass and there are no open Critical defects or unresolved Major defects affecting required SRS functionality.

9. Suspension Criteria and Resumption Requirements

- Testing is suspended if the application cannot start, repeatedly crashes before test execution, or cannot create/read its required data files.
- Testing may be suspended if a blocking defect prevents execution of a substantial portion of the planned test cases.
- Testing resumes after a corrected build is provided and smoke tests for startup/data loading, add product, and stock adjustment pass.

10. Test Deliverables

- This Test Plan.
- Test case specifications (Section 18).
- SRS-to-test-case Traceability Matrix (Section 19).
- Test execution log and defect reports.
- Test summary report.
- Test data and supporting test scripts/harnesses, if used.

11. Testing Tasks

Task	Depends On	Output
Prepare test data	Approved build / SRS	Valid products, duplicate IDs, invalid inputs, stock changes, threshold cases
Prepare test cases	SRS and RTM	TC-01 to TC-25
Execute functional tests	Build available	Execution log for TC-01 to TC-18
Execute security validation	Functional tests available	Security findings for TC-19 to TC-24
Execute boundary/summary tests	Functional tests available	Results for TC-21 to TC-25
Log defects and retest	All executions	Defect log and retest results
Prepare test summary	All tasks	Final test summary report

12. Environmental Needs

Item	Specification
Hardware	PC/laptop capable of compiling and executing the C/C++ console application
Operating system	OS supported by the chosen C/C++ compiler
Compiler	C or C++ compiler appropriate to the implementation

Storage	Local writable storage for product and stock-history files
Tools	Text/hex editor for file inspection where required; optional C/C++ test harness
Test data	Valid and invalid product records, stock-in/out values, duplicate IDs, unknown searches and low-stock threshold cases

13. Responsibilities

Member	Responsibility
Malavika A	Test coordination, product record and search test execution, defect tracking
Manoj Kumar	Stock adjustment, stock availability and history test execution
Mayank M	Security validation, negative/boundary testing and traceability verification
LG Vivyn Krishnaa	Persistence, inventory summary, restart tests and test documentation

14. Staffing and Training

All four team members participate in testing. Team members shall be familiar with the Phase-1 SRS, the console menu, the product and stock operations, and the format used to record test results. No external training is required for the planned test activities.

15. Schedule

Activity	Planned timing
Test plan and test cases ready	Before implementation/testing phase
Unit and integration testing	During implementation
System and security validation	After functional code completion
Regression testing	After defect fixes and before final submission
Test summary report	Before final project submission

16. Risks and Contingencies

Risk	Contingency
Late delivery of implementation	Execute high-risk and core FR/VFR tests first.
Team member unavailable	Use the documented test cases so another member can execute them.
File/environment issue	Use a clean test directory and backup test data; repeat after environment correction.
Many defects found late	Prioritize defects affecting core product and stock functions, then perform regression testing.

17. Approvals

Role	Name	Signature / Date
Test Lead	Malavika A	
Project Members	Malavika A; Manoj Kumar; Mayank M; LG Vivyn Krishnaa	
Course Instructor	Ambika K	

18. Test Case Specifications

The following 25 test cases cover the functional and verifiable requirements in the Phase-1 SRS. Status is filled during execution.

TC ID	Title	Req. IDs	Type	Precondition	Steps / Test Data	Expected Result	Status
TC-01	Add product with valid data	FR02	Functional	Application started; no duplicate ID	Enter unique ID, name, category, positive price and non-negative quantity.	Product is created and all entered details are stored/displayed correctly.	Not Run
TC-02	Reject duplicate product ID	FR02, VFR01	Functional Negative	One product already exists	Attempt to add another product using the same product ID.	Duplicate ID is rejected; existing record remains unchanged; no second record is created.	Not Run
TC-03	Reject missing/negative product input	FR02, VFR02	Functional Negative / Security	Add-product menu available	Try missing required field; negative unit price; negative quantity.	Each invalid entry is rejected with a validation message; no invalid record is stored.	Not Run
TC-04	Update product details	FR03	Functional	Existing product	Modify name/category/price or other supported product details with valid data; search again.	Updated details are stored and displayed; product ID remains associated with the same product.	Not Run
TC-05	Remove existing product	FR04	Functional	Existing product	Select an existing product and perform remove/delete operation.	Product is removed and is no longer returned as an active stored product.	Not Run
TC-06	Search by valid product ID	FR05, VFR03	Functional	Known product exists	Enter a valid stored product ID.	Matching product record is displayed.	Not Run
TC-07	Search with unknown key	FR05, VFR04	Functional Negative	Application contains products	Search using an unknown ID/name.	System displays a not-found result/message and does not modify data.	Not Run
TC-08	View selected product details	FR06	Functional	Known product	Select/view a product through the supported menu option.	Product details are displayed accurately.	Not Run
TC-09	Stock-in valid quantity	FR07, VFR05	Functional	Product exists with known quantity	Record a stock increase by a valid amount.	New quantity equals previous quantity plus entered amount; operation is successful.	Not Run
TC-10	Stock-out valid quantity	FR07	Functional	Product has sufficient stock	Record a stock decrease less than or equal to available quantity.	New quantity equals previous quantity minus entered amount.	Not Run
TC-11	Reject stock-out greater than available	FR07, VFR06	Functional Negative / Security	Product has known available quantity	Attempt stock-out greater than available quantity.	Operation is rejected; available quantity remains unchanged.	Not Run

Retail Inventory and Stock Management System - Test Plan | Team E10

TC-12	Check current stock availability	FR08	Functional	Product exists	Select product and use stock availability option.	Displayed quantity equals the stored current quantity.	Not Run
TC-13	Record successful stock adjustment	FR09, VFR07	Functional	Successful stock-in/out completed	Open stock movement history file or supported history view.	Each successful adjustment contains product ID, quantity change, date/time and adjustment type.	Not Run
TC-14	View stock movement history	FR10	Functional	Product has one or more stock changes	Request history for the product.	History for the selected product is displayed correctly and corresponds to recorded adjustments.	Not Run
TC-15	View all product records	FR11	Functional	Multiple products stored	Select inventory/list option.	All stored product records are listed with their relevant details.	Not Run
TC-16	Inventory summary and low-stock boundary	FR12, VFR09	Functional / Boundary	Low-stock threshold is configured	Create/adjust products to quantities below, equal to and above the threshold; generate summary.	Products at or below the configured threshold are identified as low-stock; others are not.	Not Run
TC-17	Product and history persistence	FR13, VFR08	Functional / Reliability	Valid records and stock history exist	Exit application normally; restart; inspect products and history.	Previously stored product and stock-history data remains available and unchanged.	Not Run
TC-18	Load saved data at startup	FR14, VFR08	Functional	Saved data exists before launch	Start the application with existing files.	Previously stored product and stock-history data is loaded and usable after startup.	Not Run
TC-19	Authorized user access	FR01	Security Validation	Application access mechanism implemented	Enter the valid authorized-user input required by the implementation; attempt an invalid access if supported.	Authorized user can enter the application; invalid/unauthorized access is rejected where an access mechanism exists.	Not Run
TC-20	Access boundary / no unintended menu bypass	FR01	Security Validation	Application started at access point	Attempt to reach protected inventory operations without completing the required access step, if access control is implemented.	System does not expose protected operations before required access; behavior matches implemented access design.	Not Run
TC-21	Malformed and extreme product input	VFR02	Security / Boundary	Add-product prompts available	Enter empty fields, very long text, special characters, negative price/quantity and other invalid values supported	Program does not crash or hang; invalid input is rejected; files remain intact.	Not Run

					by the input format.		
TC-22	Invalid search and stock input	VFR04, VFR06	Negative / Boundary	Search and stock menus available	Use malformed/unknown search keys and stock-out values greater than available quantity.	Invalid operation is rejected; no unintended data change occurs.	Not Run
TC-23	Stock quantity calculation boundary	VFR05, VFR06	Boundary	Known starting quantity	Perform stock-in of a small valid quantity, stock-out up to available quantity, then attempt one unit beyond available.	Valid operations calculate correctly; excessive stock-out is rejected.	Not Run
TC-24	Rejected operations do not alter stored history	FR09, FR13, VFR07, VFR08	Security / Persistence	Known product and history	Record a valid adjustment, then perform rejected duplicate/invalid product and stock operations; inspect/reload history.	Only successful stock adjustments are recorded; rejected operations do not create false history entries; previous records persist.	Not Run
TC-25	Low-stock threshold verification	FR12, VFR09	Boundary	Configured threshold known	Set quantity to threshold+1, threshold, and threshold-1; generate inventory summary for each state.	Only quantities at or below the configured threshold are identified as low-stock.	Not Run

19. Traceability Matrix (SRS to Test Cases)

Every FR01–FR14 and VFR01–VFR09 in the supplied SRS is covered by at least one test case. The mapping below provides the required SRS traceability aspect.

Requirement	Category	Test Cases	Count
FR01	Functional	TC-19, TC-20	2
FR02	Functional	TC-01, TC-02, TC-03	3
FR03	Functional	TC-04	1
FR04	Functional	TC-05	1
FR05	Functional	TC-06, TC-07	2
FR06	Functional	TC-08	1
FR07	Functional	TC-09, TC-10, TC-11, TC-23	4
FR08	Functional	TC-12	1
FR09	Functional	TC-13, TC-24	2
FR10	Functional	TC-14	1
FR11	Functional	TC-15	1
FR12	Functional	TC-16, TC-25	2
FR13	Functional	TC-17, TC-24	2
FR14	Functional	TC-18	1
VFR01	Verifiable Functional	TC-02, TC-21	2
VFR02	Verifiable Functional	TC-03, TC-21	2
VFR03	Verifiable Functional	TC-06	1
VFR04	Verifiable Functional	TC-07, TC-22	2
VFR05	Verifiable Functional	TC-09, TC-23	2
VFR06	Verifiable Functional	TC-11, TC-22, TC-23	3
VFR07	Verifiable Functional	TC-13, TC-24	2
VFR08	Verifiable Functional	TC-17, TC-18, TC-24	3
VFR09	Verifiable Functional	TC-16, TC-25	2

Coverage statement: FR01–FR14 and VFR01–VFR09 are each linked to one or more planned test cases. Test execution results shall be recorded against the same TC IDs so requirement coverage can be reported after execution.